

MSE 311 Thermodynamics of materials-Homework 1

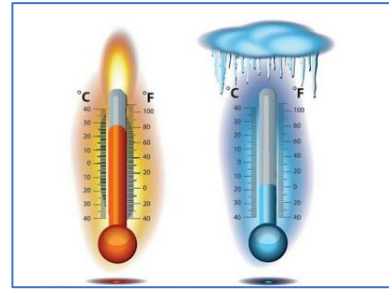
Student ID:

Name:

1. Please prove the following formula according to the *entropy increase equation*.

$$\eta = \frac{W}{Q_H} = \frac{Q_H - Q_C}{Q_H} \leq \frac{T_H - T_C}{T_H}$$

2. Whether the temperature scale of mercurial thermometer should be uniform or not? Please does derivation considering the coefficient of *linear thermal expansion* (α) is constant: is the volume increment of mercury caused by temperature rise linearly related to the temperature difference? Assuming the length of the material is L_0 , the length increases ΔL when temperature rises ΔT . Then α can be calculated by follows.



$$\alpha = \frac{\Delta L}{L_0 \Delta T}$$